# Software Design Analysis — Quiz

**Total Marks: 20 | Time: 30 minutes**

## Section A: Fill in the Blanks (10 Marks — 1 mark each)

**Q1.** Homogenization of classes is the process of ***grouping similar classes*** with ***shared*** characteristics into a common superclass to reduce design redundancy.

**Q2.** In a component diagram, a ***required (lollipop-and-socket / "usage")*** interface represents a service that a component requires from its environment to function.

**Q3.** A deployment diagram shows the physical allocation of ***artifacts / software components*** to ***hardware*** nodes in a system's runtime environment.

**Q4.** In a state chart diagram, a ***composite*** state that contains nested sub-states within it.

**Q5.** In a deployment diagram, when the same software artifact is deployed on multiple nodes to handle load, this architecture reflects the ***scalability and availability*** quality attributes

**Q6.** In component diagrams, components communicate with each other through well-defined ***interfaces*** which hide internal implementation details.

**Q7.** In a statechart, a ***transition (trigger)*** is a condition or event that causes the system to move from one state to another.

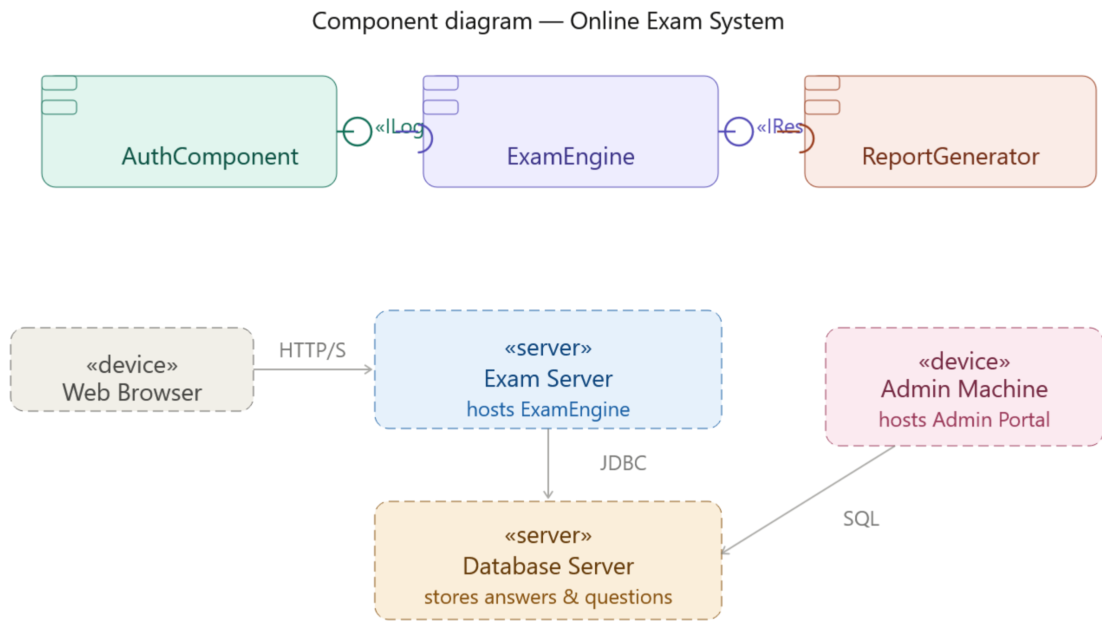
**Q8.** In a component diagram, when Component A has a required interface that matches Component B's provided interface, the connection between them is called a(n) ***assembly connector.***

**Q9**. A state that is active as long as no event occurs and automatically transitions after a fixed duration uses a special trigger called a(n) ***time / after(duration).***

**Q10.** In a deployment diagram, a ***node*** represents a physical or virtual machine that can host and execute software components.

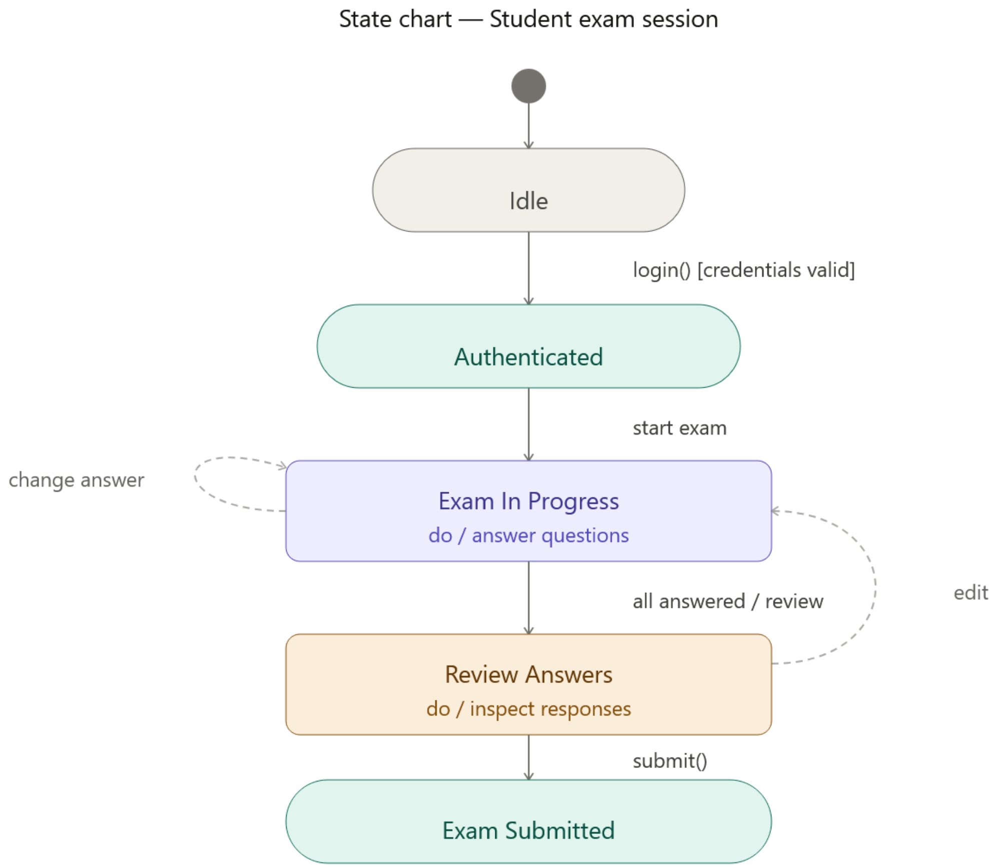
## Section B: Scenario-Based Question (10 Marks)

### Section B — Component Diagram (5 marks):




| Criteria | Marks |
|---|---|
| Correct use of component notation (box + stereotype) | 1 |
| Provided & required interfaces shown (lollipop / socket) | 1 |
| Dependency/usage relationships between components correct | 1 |
| Nodes (Browser, Exam Server, DB, Admin) shown with deployment | 1 |
| Communication paths between nodes labeled | 1 |

**Section B — State Chart Diagram (5 marks):**




| Criteria | Marks |
|---|---|
| Initial pseudo-state and at least 4 valid states | 1 |
| Transitions labeled with events/guards | 1 |
| At least one guard condition shown (e.g., `[credentials valid]`) | 1 |
| Internal activity in a state (`do /` action) | 1 |
| Logical flow correct (Idle → Auth → In Progress → Review → Submitted) | 1 |